

SAP Cloud Platform Extension Factory, Kyma Runtime

Открытая платформа SAP для расширения приложений на Kubernetes



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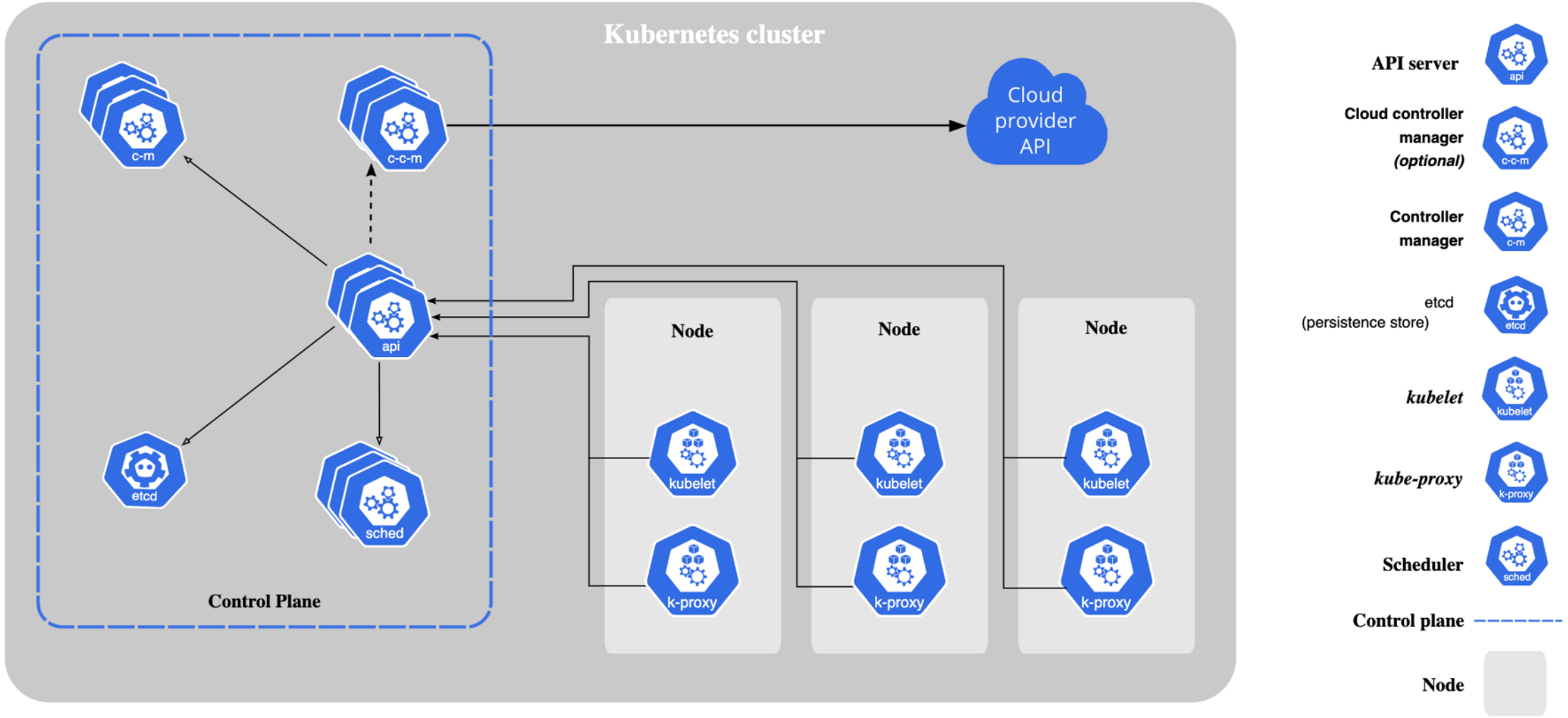
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Немного о Kubernetes

Кластер Kubernetes

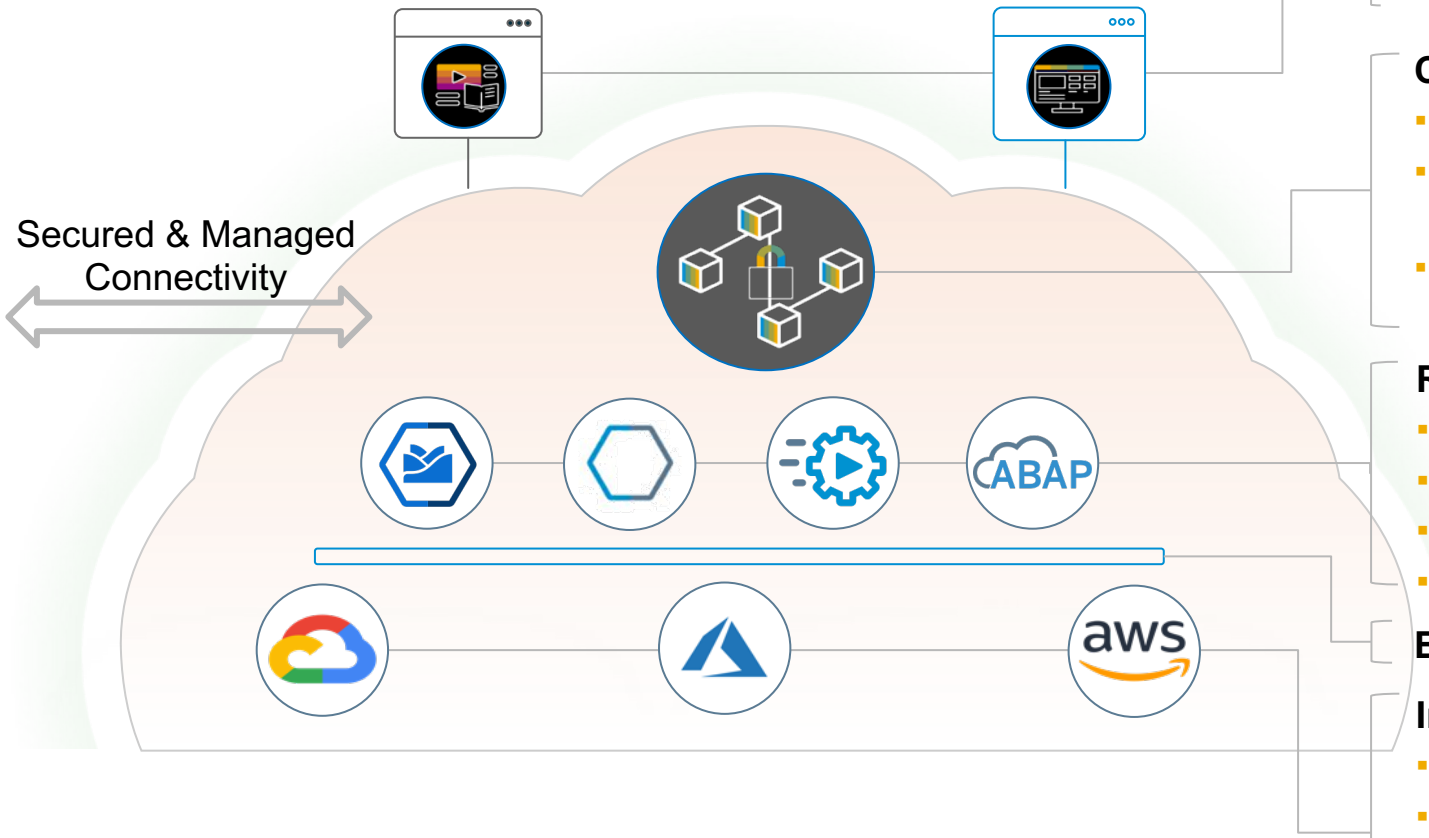


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Платформа для разработки
SAP Cloud Platform Extension Factory

SAP Cloud Platform Extension Factory

- SAP SuccessFactors
- SAP S/4HANA Cloud + On Premise [1]
- SAP C/4HANA
- SAP Fieldglass [1]
- SAP Concur [1]
- SAP Ariba [1]
- SAP Enterprise Core Components [1]
- SAP Graph [1]
- 3rd Party Solutions [1]



Multiple Administration UIs
- SAP Cloud Platform Cockpit
- LoB Admin Cockpits [1], e.g. SAP SuccessFactors Extension Center

Central Management Plane
- Secured App Connectivity
- Central Apps APIs & Events Registry
- Standardized Service Mgmt & Consumption

Runtime of choice
- Kyma runtime
- Serverless runtime
- Application runtime (CF)
- ABAP runtime

Business Event Bus

Infrastructure of choice
- Google Cloud Platform
- Amazon Web Services
- Microsoft Azure
- Alibaba Cloud
- SAP and others

1 Planned innovation

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Развертывание приложений в SCP в различных окружениях по выбору



Application Runtime (based on Cloud Foundry)

Приложения запускаются в модульном и легком контейнере

- Создание и запуск cloud-native приложений
- Применение открытых стандартов, пакеты расширений на Java и Node.js



Kyma Runtime (based on Kubernetes)

Создание облачных расширений на основе Kubernetes

- Полностью управляемая среда выполнения Kubernetes
- Построен с использованием Kyma Open Source
- Комбинация микросервисных и безсерверных (функции) расширений
- Расширения на основе событий (event-based) и сервисов (API-based)



Serverless Runtime

Создание и запуск облачных расширений в безсерверной среде

- Создание проектов расширений
- Создание безсерверных функций, которые могут запускаться по требованию
- Запуск облачных расширений в полностью управляемой среде
- Специальное приложение для управлениям функциями - Центр расширений



ABAP Environment

Запуск кастомного ABAP кода на отдельной платформе отделенной от цифрового ядра

- Повторно использовать существующие ноу-хау ABAP
- Отделение ABAP-приложения от цифрового ядра
- Масштаб с учетом потребностей бизнеса

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Для чего нужно Kyma Runtime

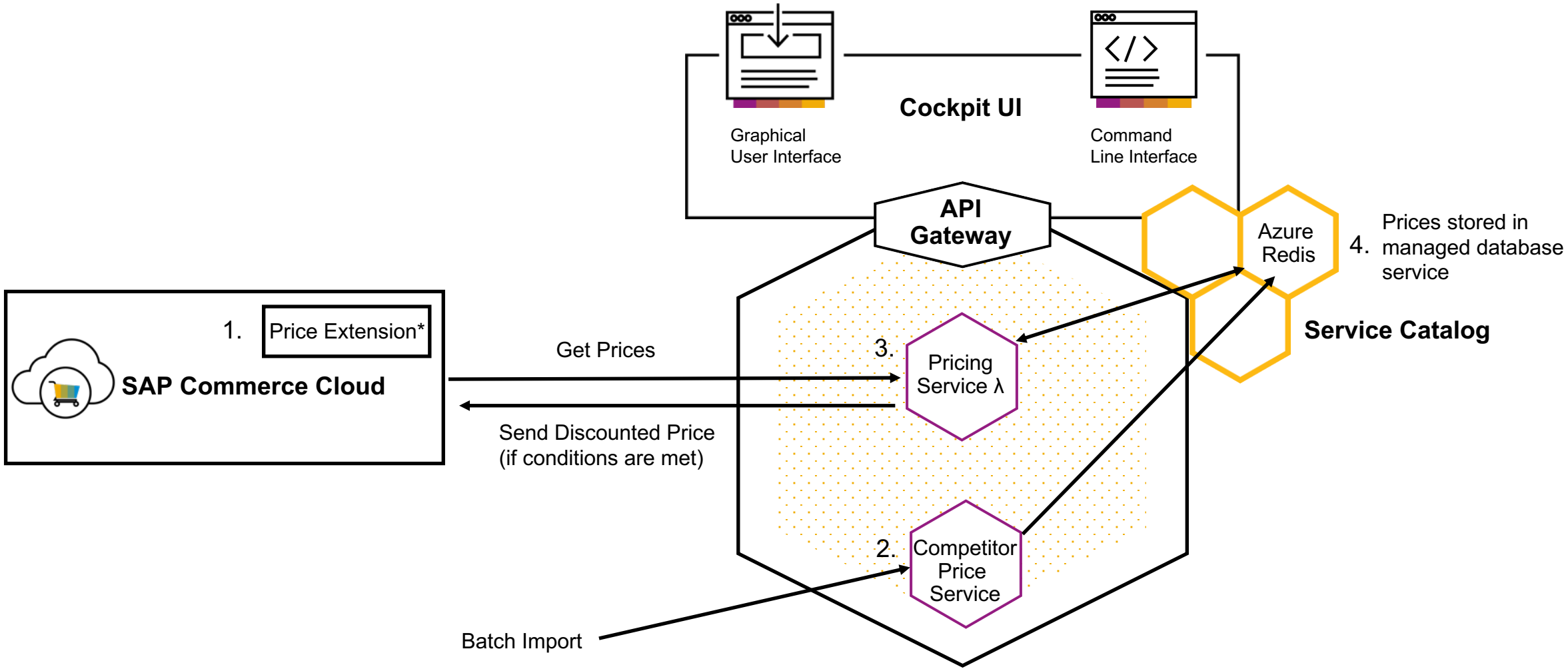
Основная цель Kyma Runtime

Enable the
Cloud-Native Developer



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Пример расширения – Ценовые предложения с SAP Cloud Platform Extension Factory и SAP Commerce Cloud



*Please note that this functionality is currently not available by standard

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Deep dive Kyma Runtime

Что внутри Kyma runtime – основные компоненты

Service Mesh: Service-to-service взаимодействие и проксирование запросов

Serverless engine: Встроенный бессерверный runtime

Event Bus: Шина событий для реагирование на бизнес-события приложений

API Gateway: Безопасный внешний доступ к API микросервисов и функций

Service Catalog: Быстрое создание и потребление сервисов

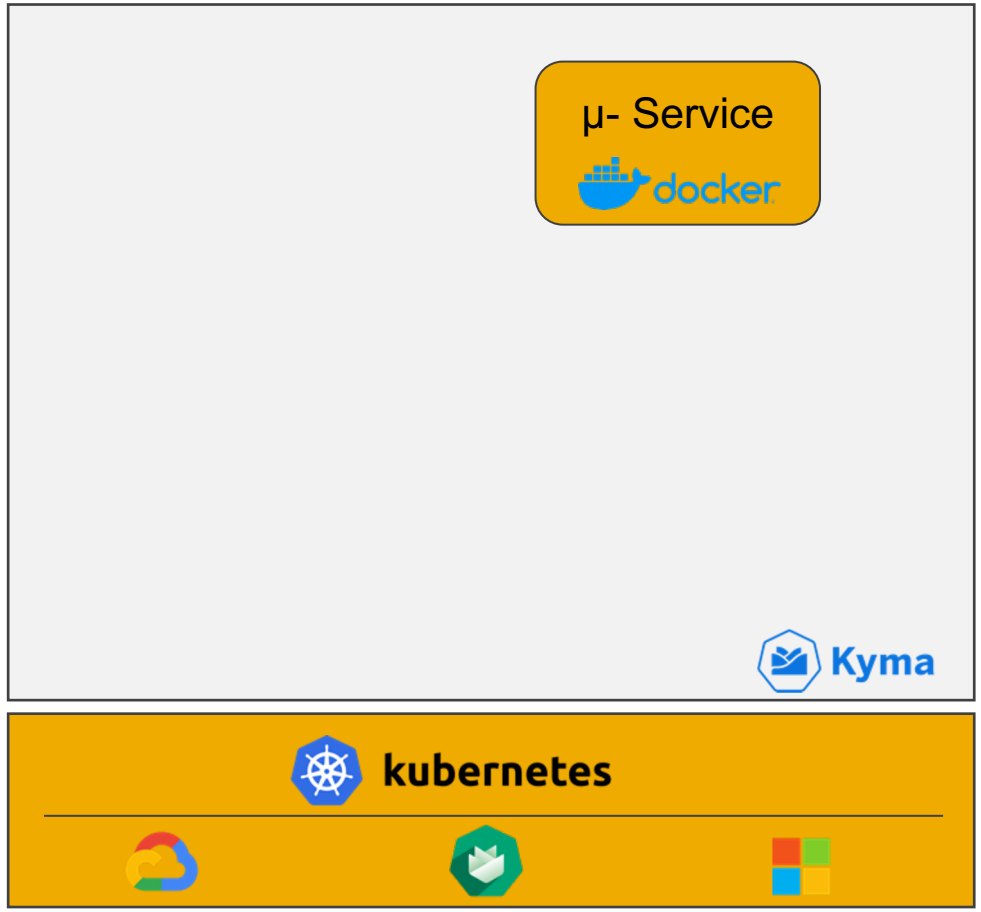
Application Connector: Безопасное подключение приложений SAP

Используются различные open source проекты



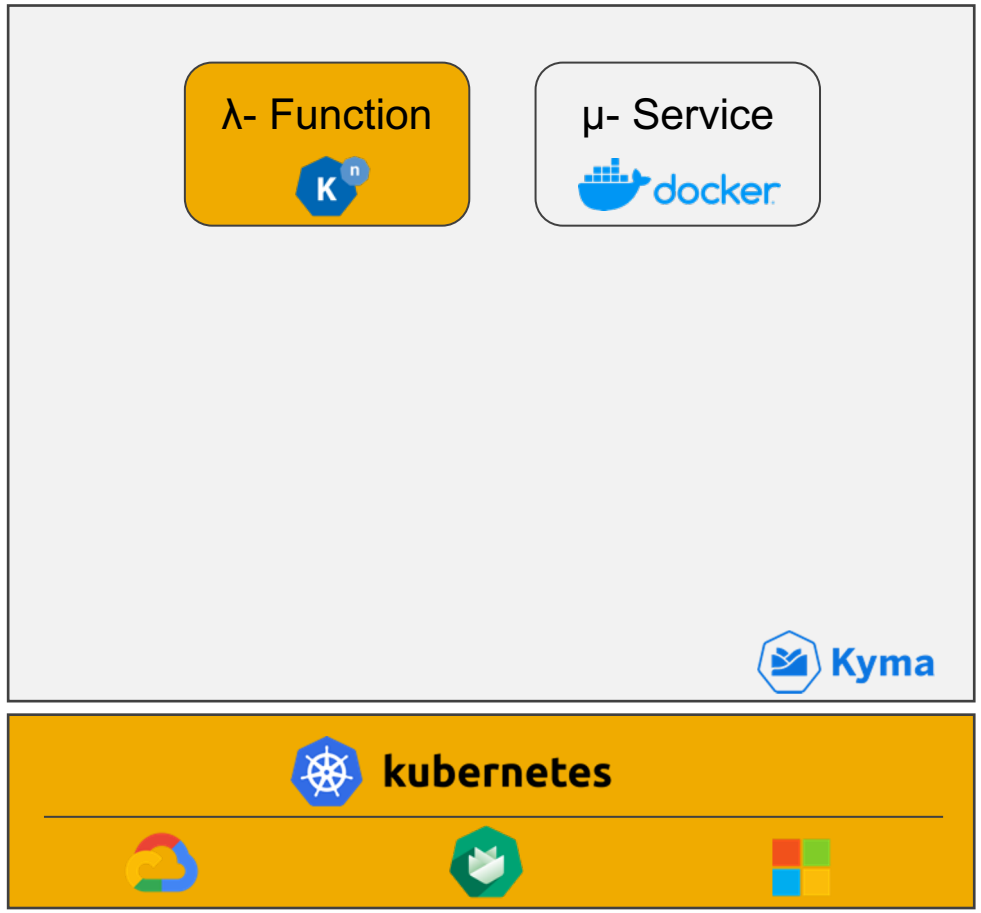
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Kyma Runtime: Разработка расширений (μ- микросервисы)



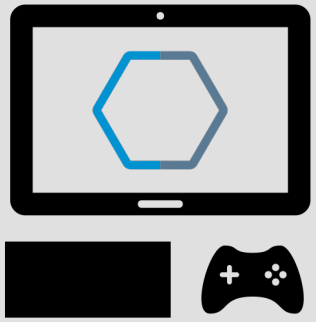
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Kyma Runtime: Serverless (λ-) функции

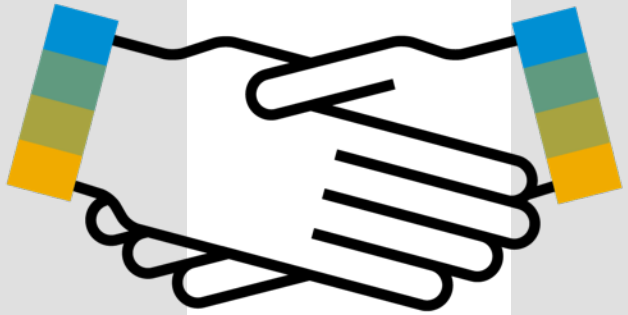


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Function-as-a-Service с SAP Cloud Platform



Serverless Runtime



Kyma Runtime



Предназначен для создания расширений на Cloud Platform

Полностью управляемый сервис, самый простой способ.

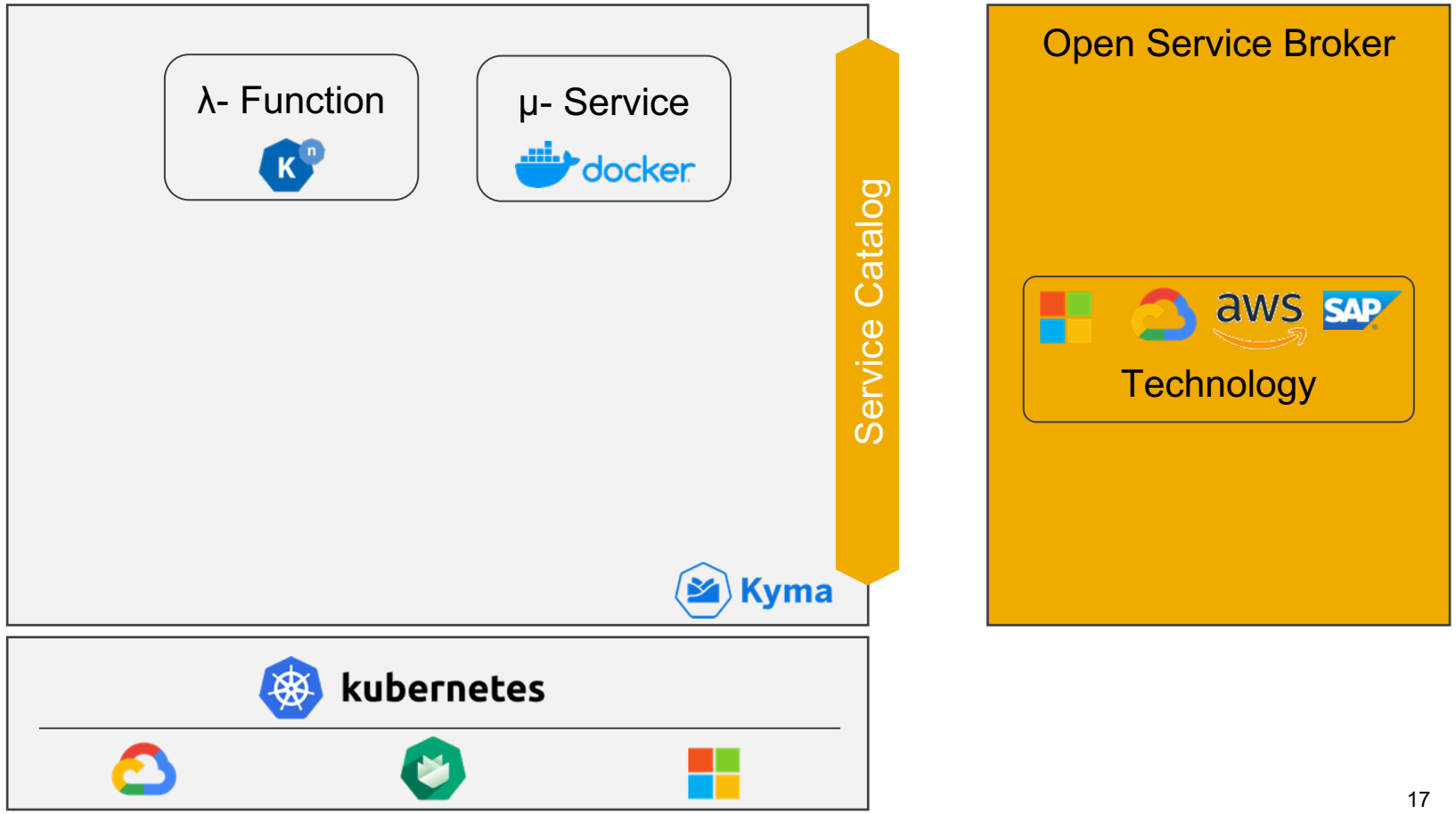
Скрытие базового технологического слоя.

Предназначен для **объединения функций и микросервисов** для построения расширений на Cloud Platform

Большая гибкость и свобода за счет доступа к технологическому слою приводит и к большей ответственности за затраты и операции.

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Kyma Runtime: Сервисы & Привязка

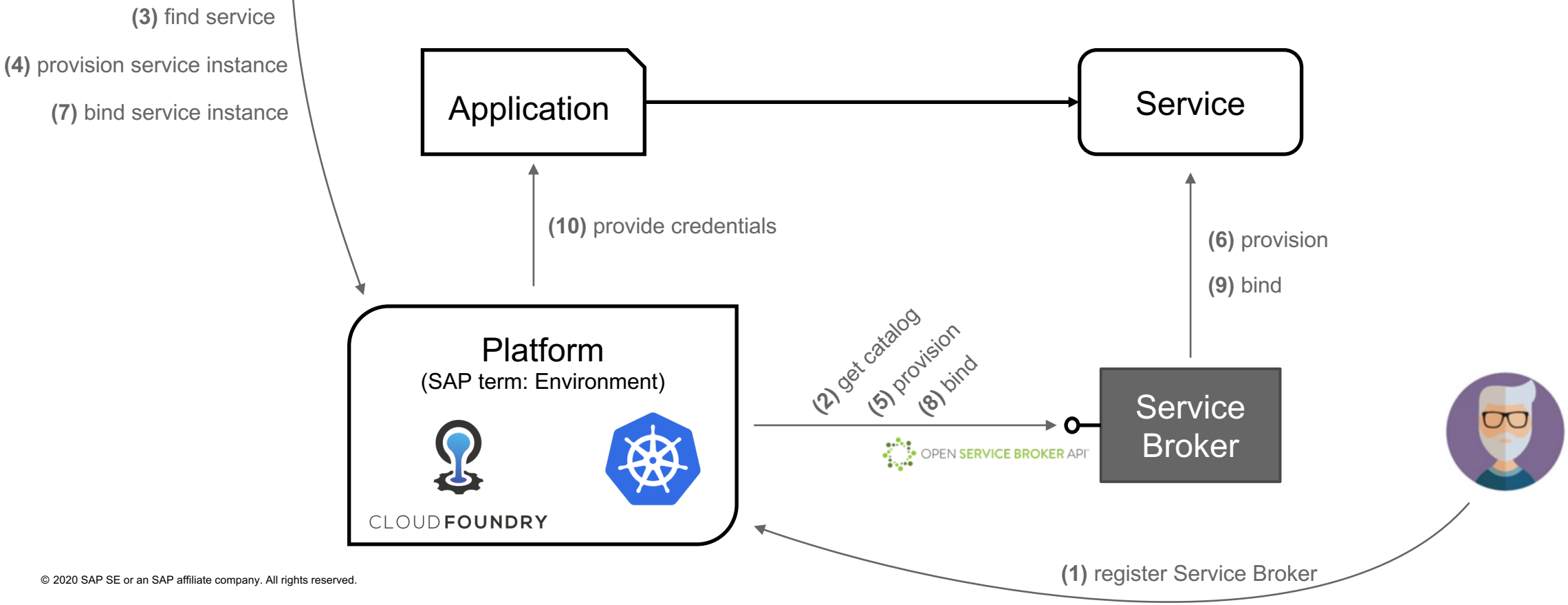


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Схема работы Open Service Broker API

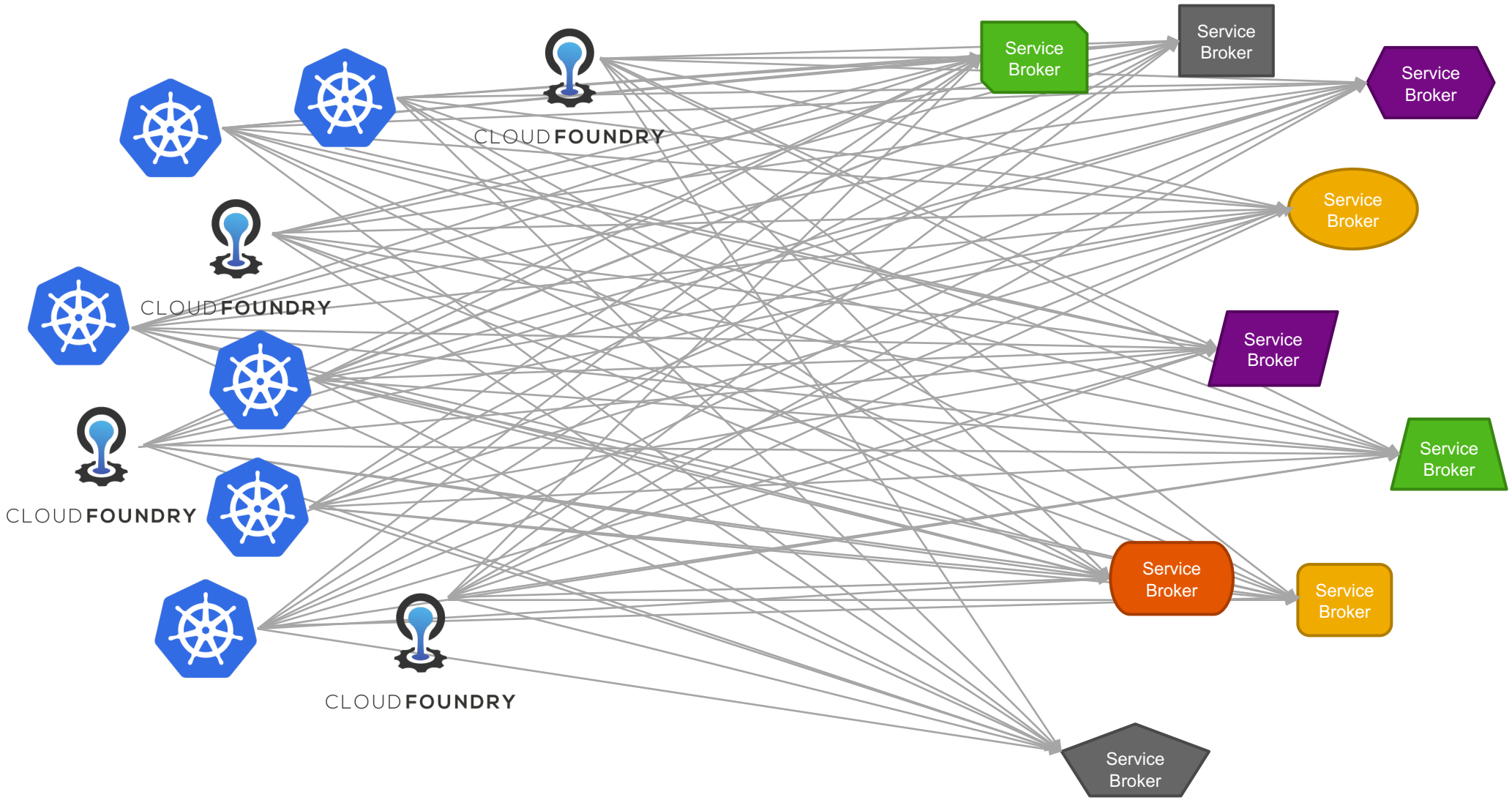
1. “Discover” сервис
2. “Provision” инстанса сервиса
3. “Bind” инстанса сервиса к приложение и получение прав (”credentials”)





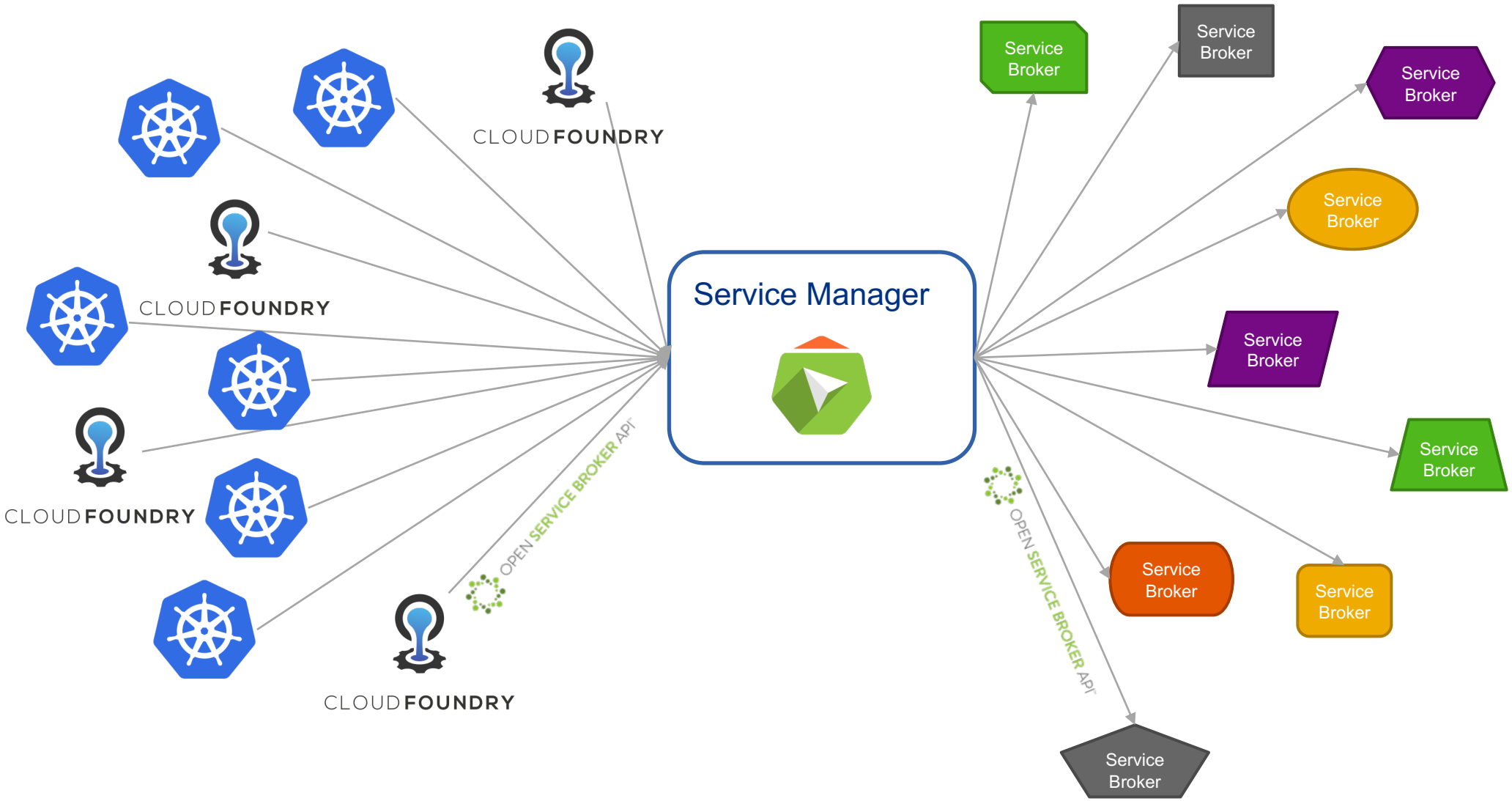
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Управление сервисными брокерами в нескольких средах



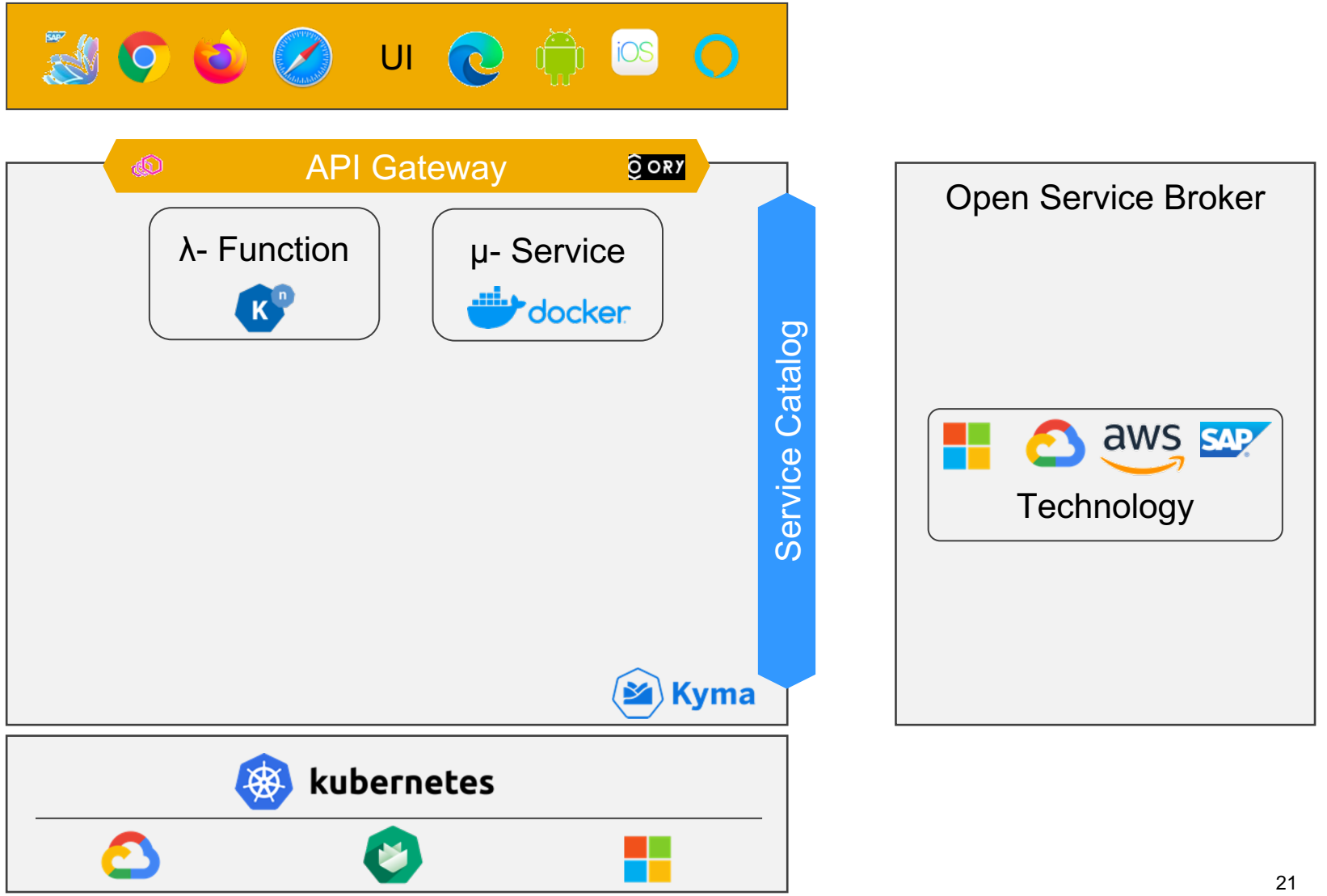
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Service Manager



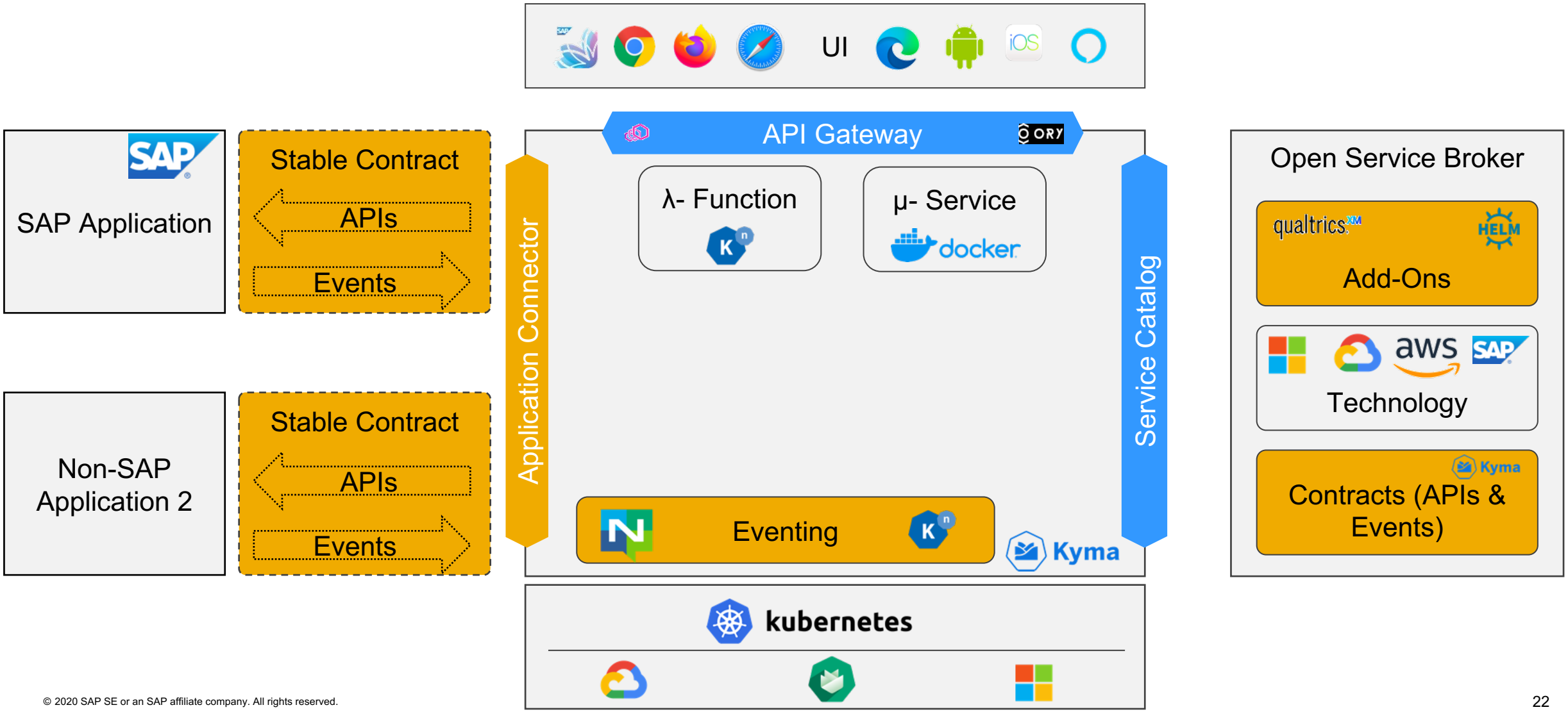
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Kyma Runtime: Фронтенд



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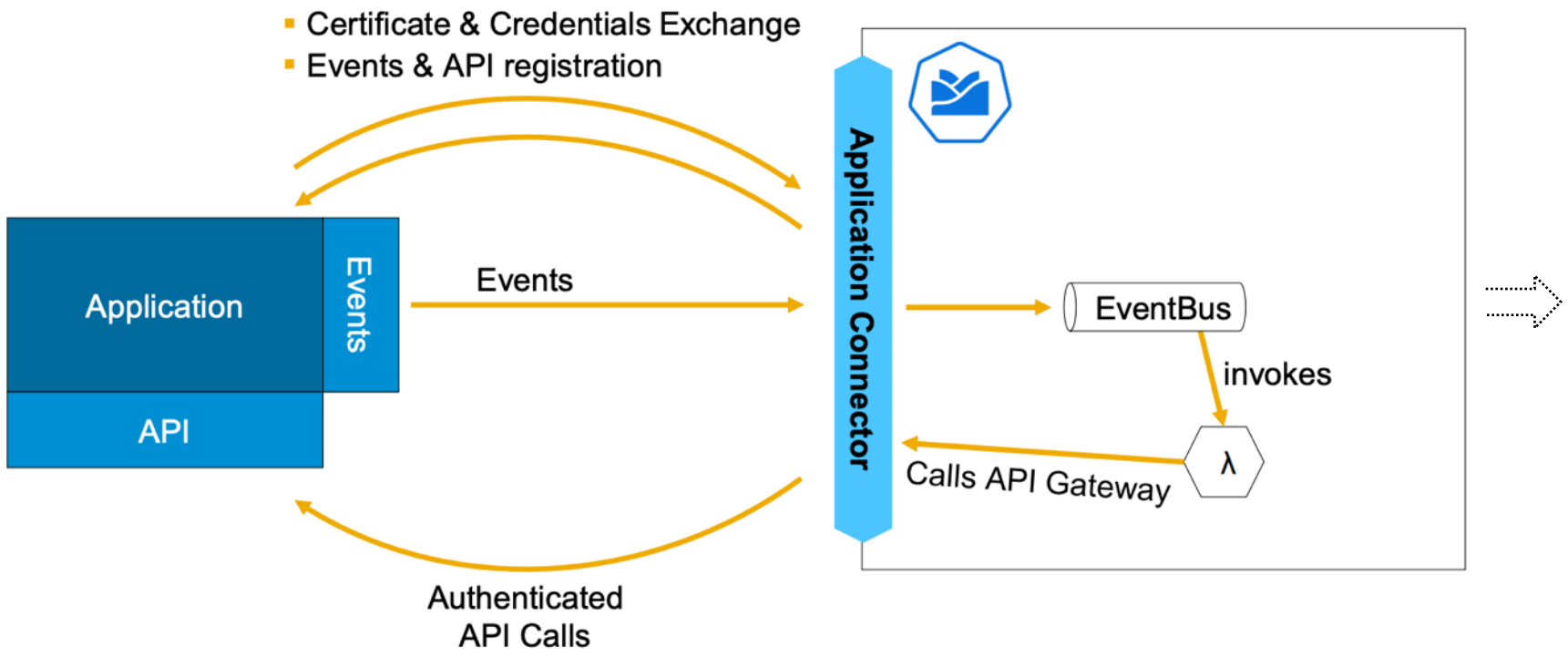
Kyma Runtime: Application Connectivity



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Kyma Runtime: Application Connectivity

Варианты коннекторов

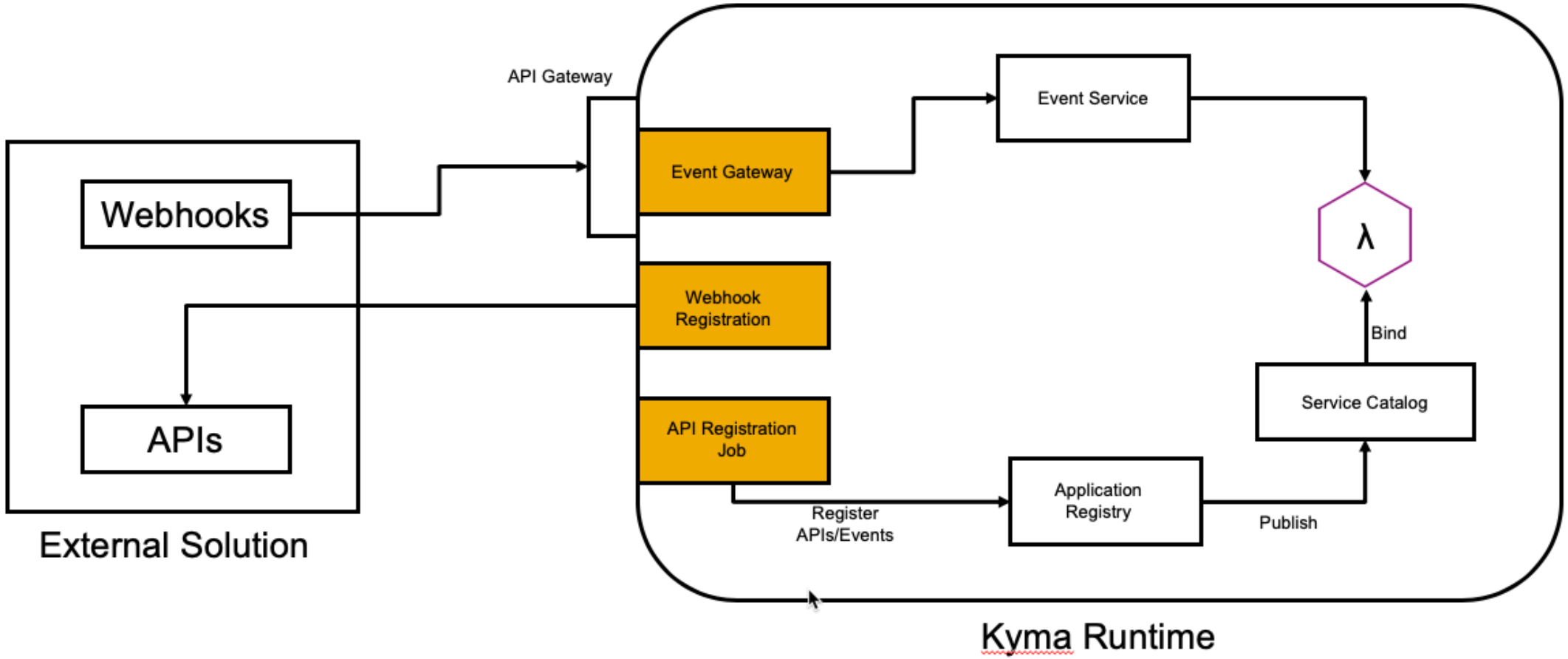


1. **Application connectors**
2. **Add-ons**

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Kyma Runtime: Application Connectivity

Структура и схема работы add-ons

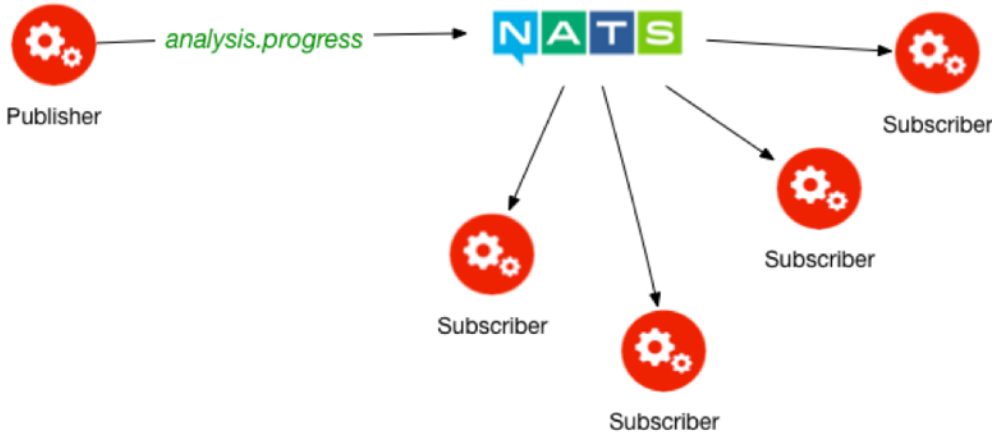


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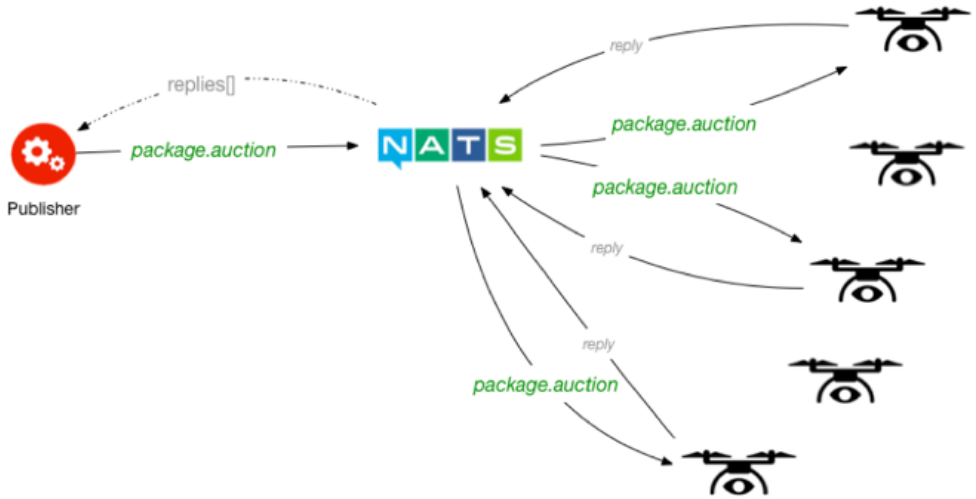
Kyma Runtime: Application Connectivity

Обработка событий через NATS messaging

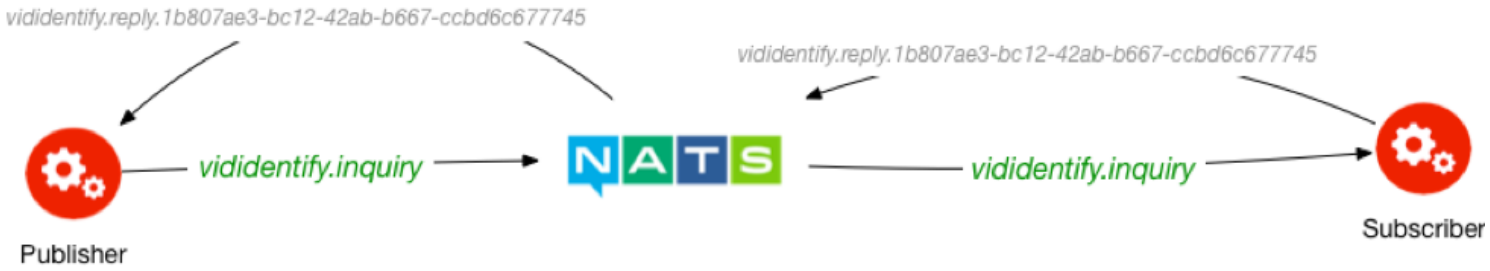
Publish and Subscribe



Scatter-Gather

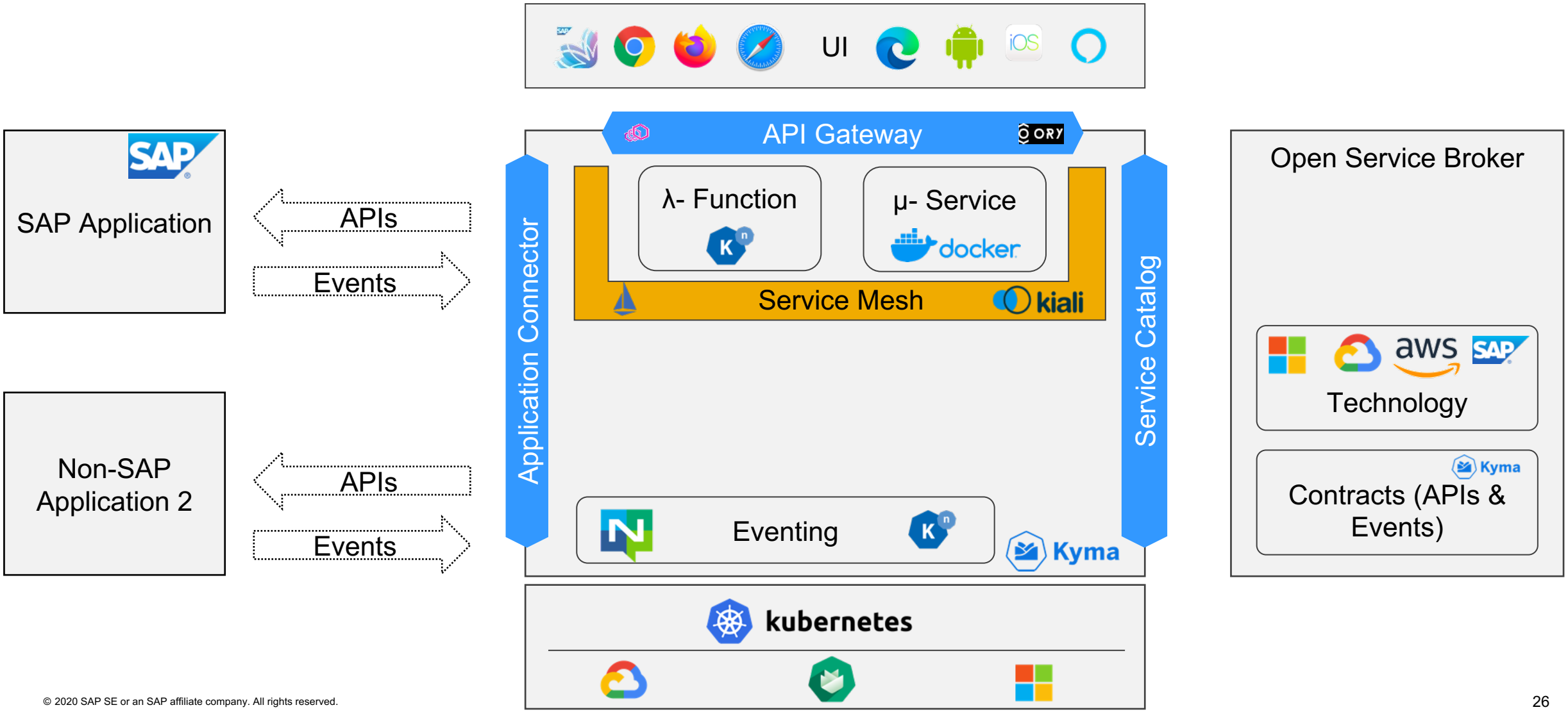


Request-Reply



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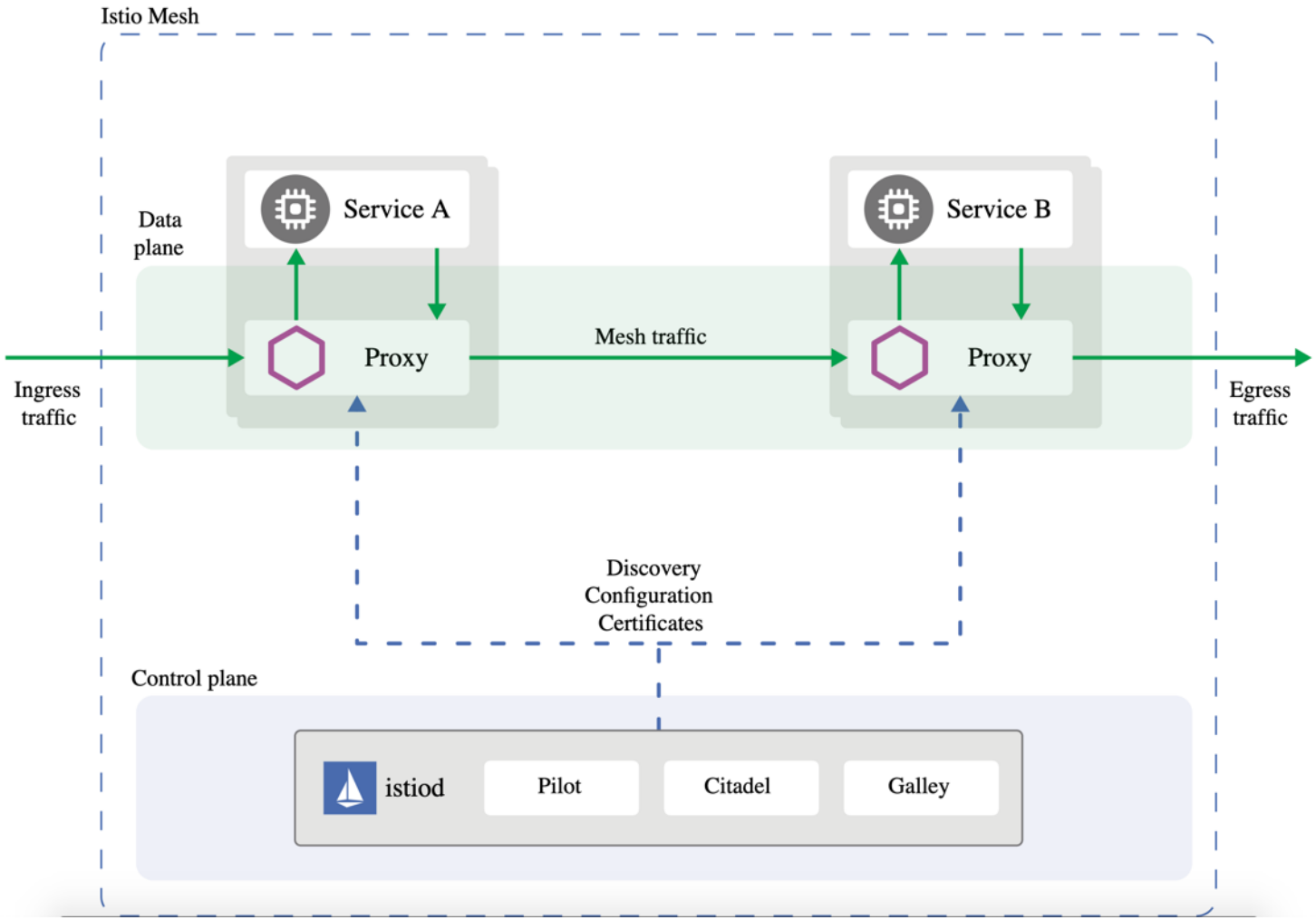
Kyma Runtime: Управление микросервисами



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Kyma Runtime: Управление микросервисами

Схема работы Istio service mesh



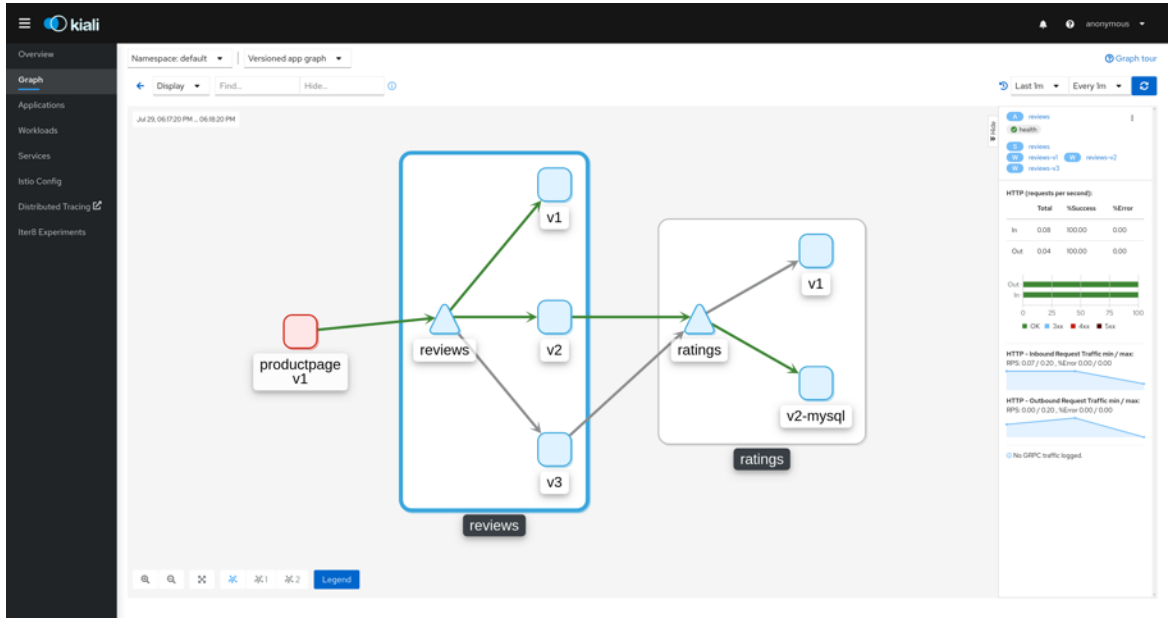
Source: https://istio.io/latest/docs/concepts/what-is-istio/

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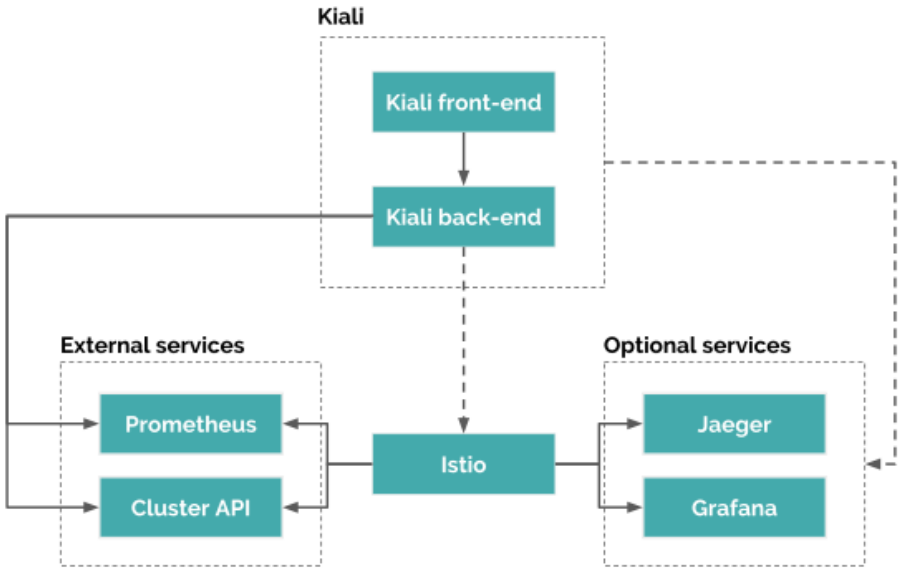
Kyma Runtime: Управление микросервисами

Визуализация service mesh через Kiali

Графы микросервисов



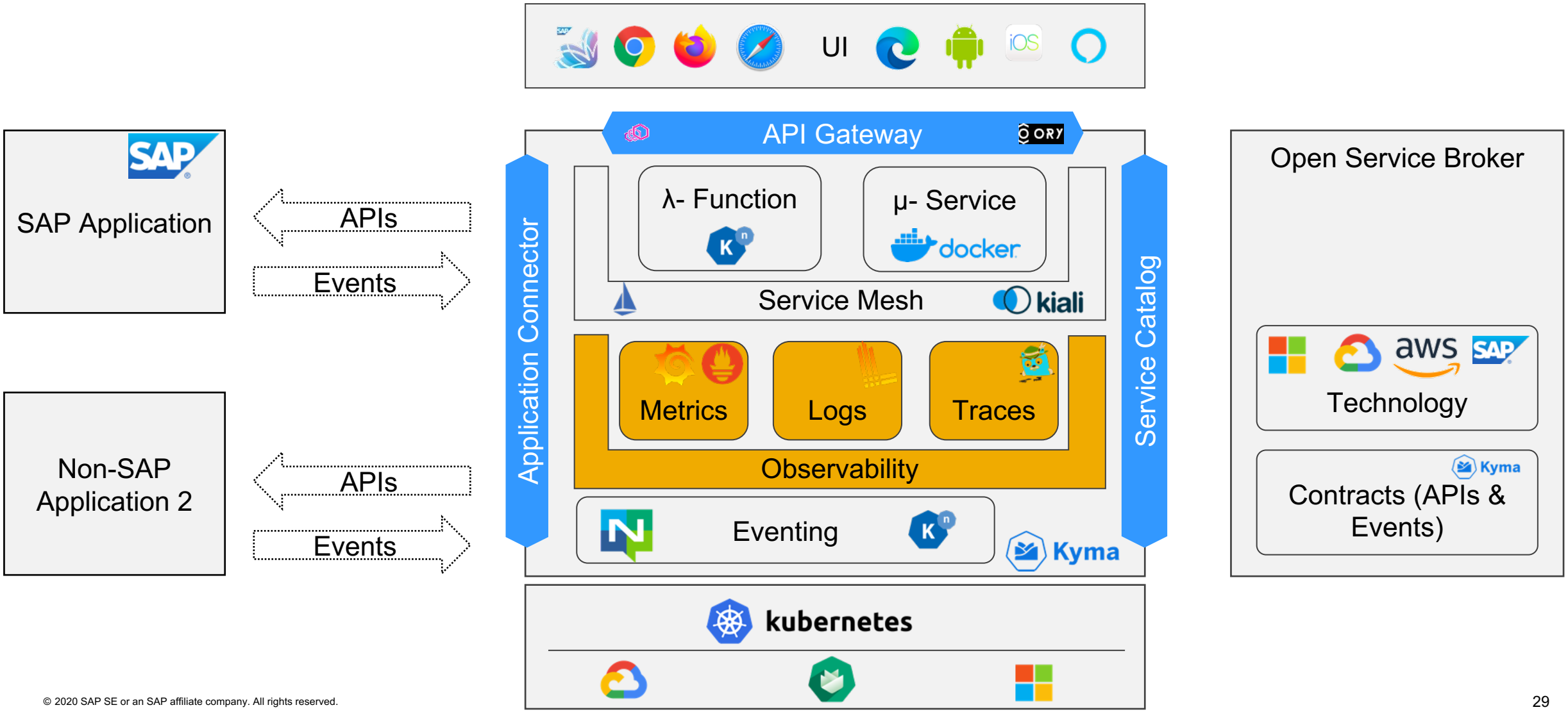
Архитектура Kiali



Source: https://kiali.io/documentation

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Kyma Runtime: Логирование и мониторинг

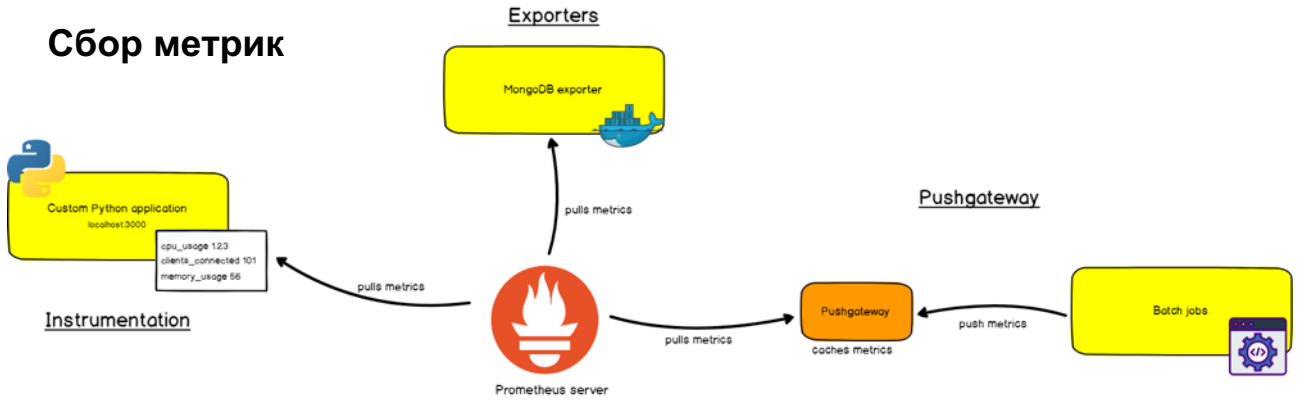


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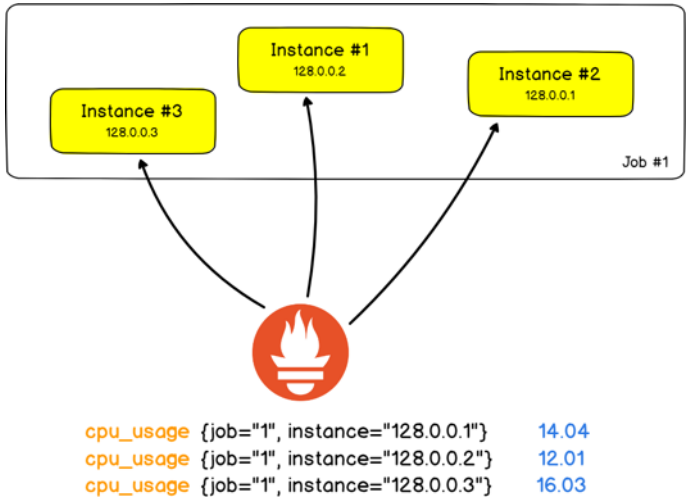
Kyma Runtime: Логирование и мониторинг

Сбор метрик посредством Prometheus

Сбор метрик



Модель данных



Оповещения

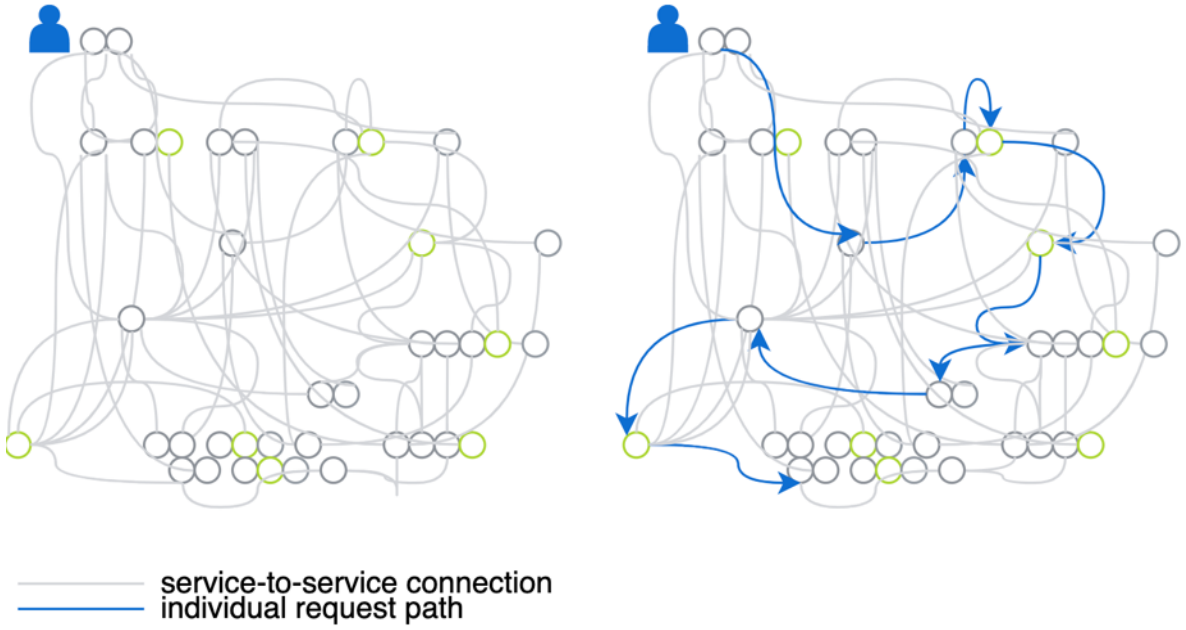


Source: https://devconnected.com/the-definitive-guide-to-prometheus-in-2019/

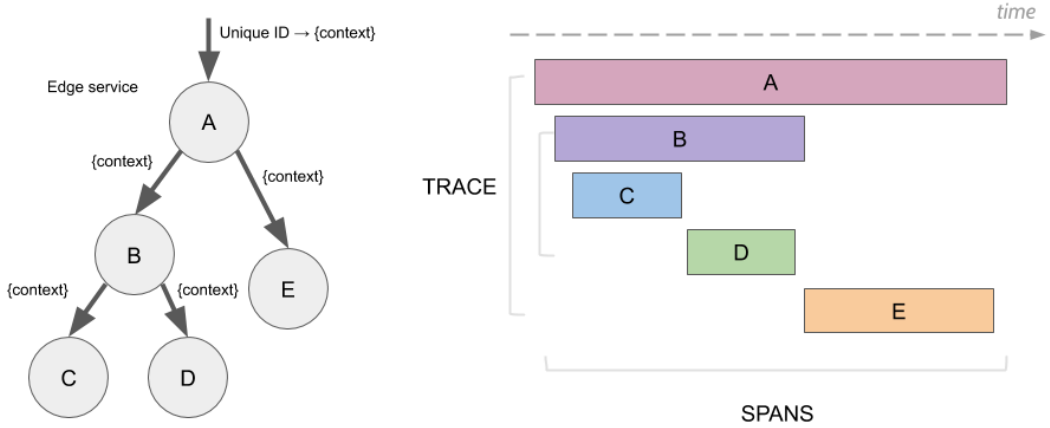
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Схема работы трассировки сервиса (Jaeger)

Распределенная трассировка микросервисов



OpenTracing Specification

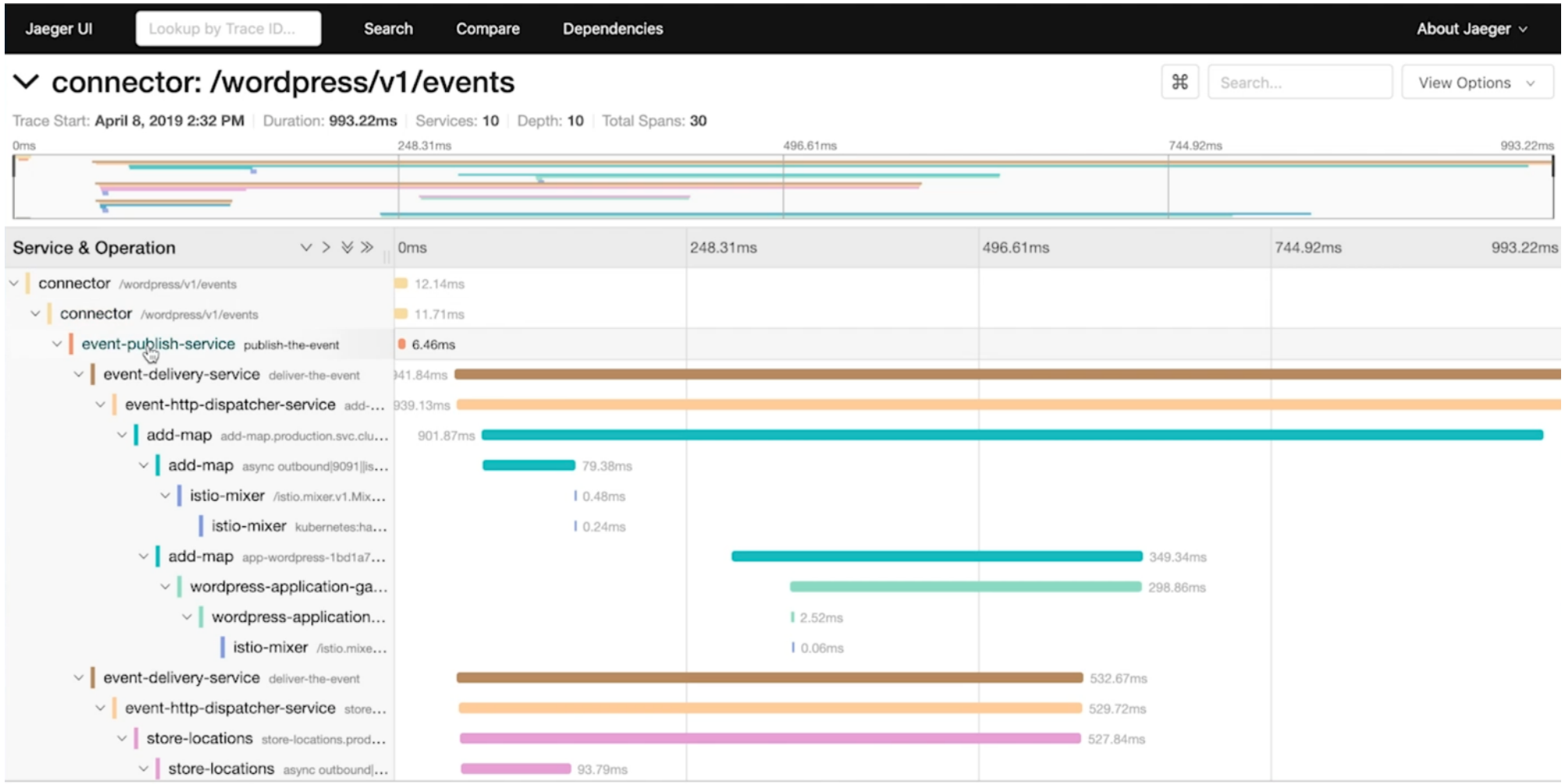


Source: https://www.jaegertracing.io/docs/1.19/architecture

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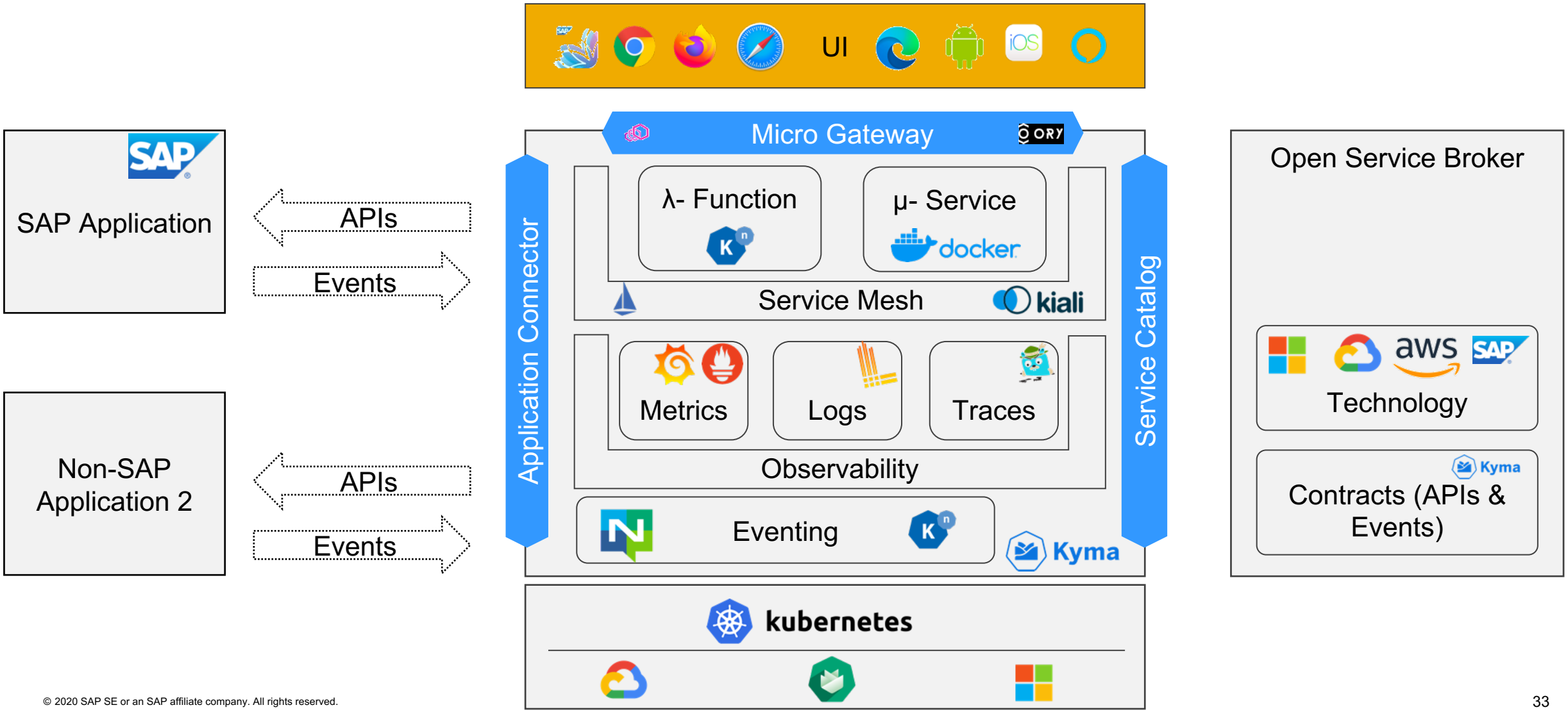
Kyma Runtime: Логирование и мониторинг

Схема работы трассировки сервиса (Jaeger)



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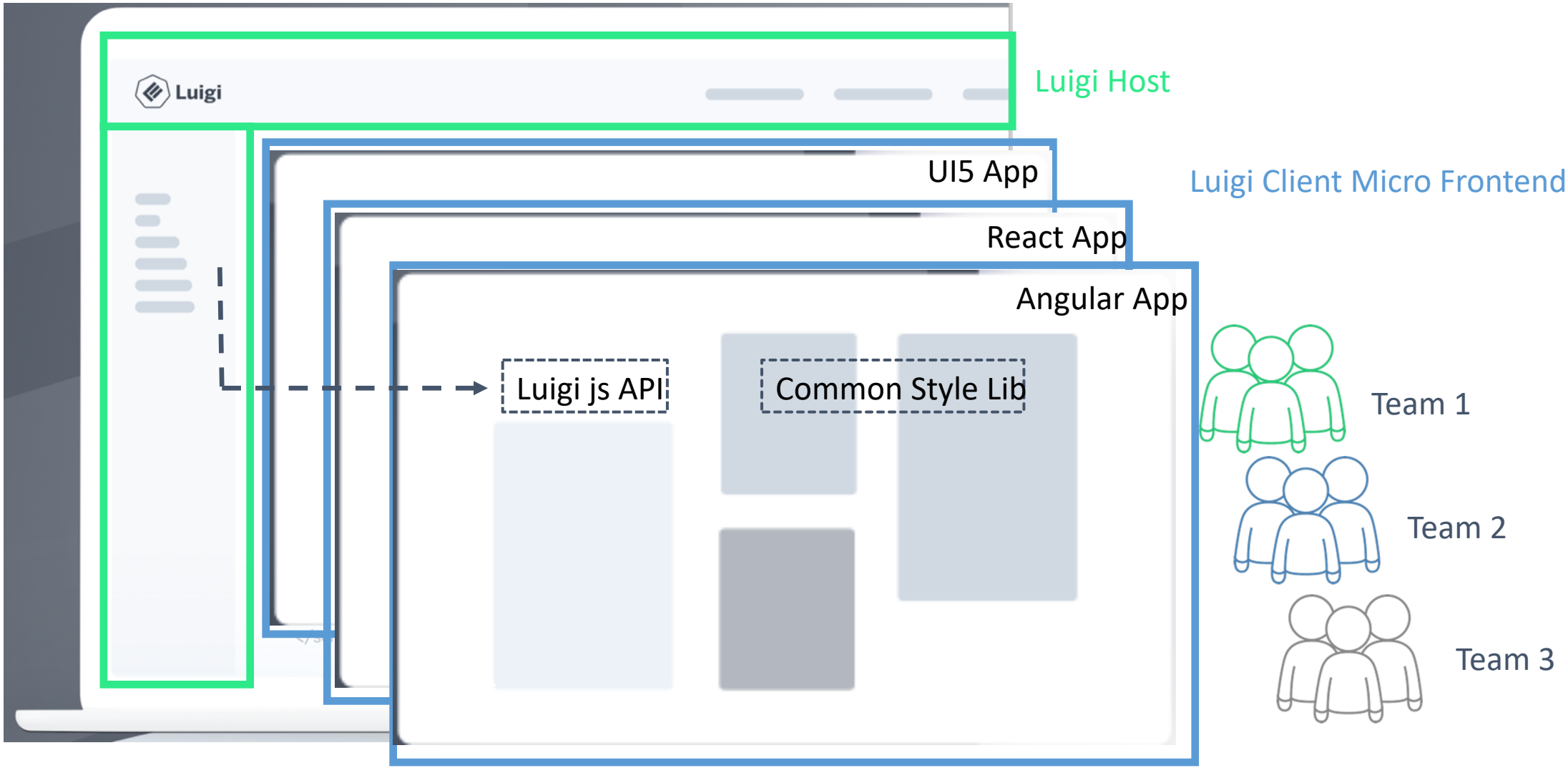
Kyma Runtime: User Interface



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Kyma Runtime: User Interface

Luigi – microfrontend frameworks



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Разработка расширений различных типов в Kyma Runtime

Запуск	Основанное на событиях (Event-based) ▪ Расширение триггера по бизнес-событиям ▪ Используется для асинхронных процессов	Основанное на (API-based) ▪ Публикация API для вызова расширений ▪ Используется для синхронных процессов и расширения пользовательского интерфейса.
Тип	Serverless функции ▪ Использование встроенного serverless движка ▪ Построение чистых (pure) функций-расширений	Микросервисы ▪ Развертывание расширений в контейнерах ▪ Использовать любой язык программирования

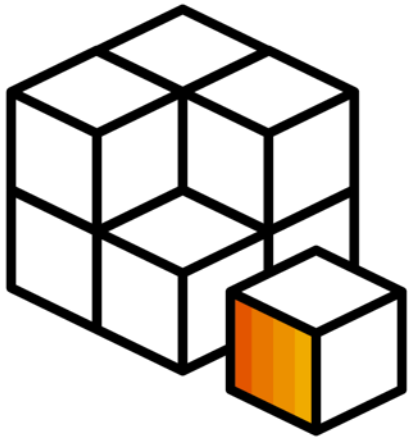
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Демонстрация

Kyma runtime является Cloud-Native



Стек основан на компонентах и стандартах с открытым исходным кодом



Обеспечивает развертывание высокомасштабируемых приложений в архитектуре контейнерных микросервисов



Использование лучших практик Производительности, Масштабируемости, Эффективности, Безопасности при разработки приложений

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Kyma в интернете

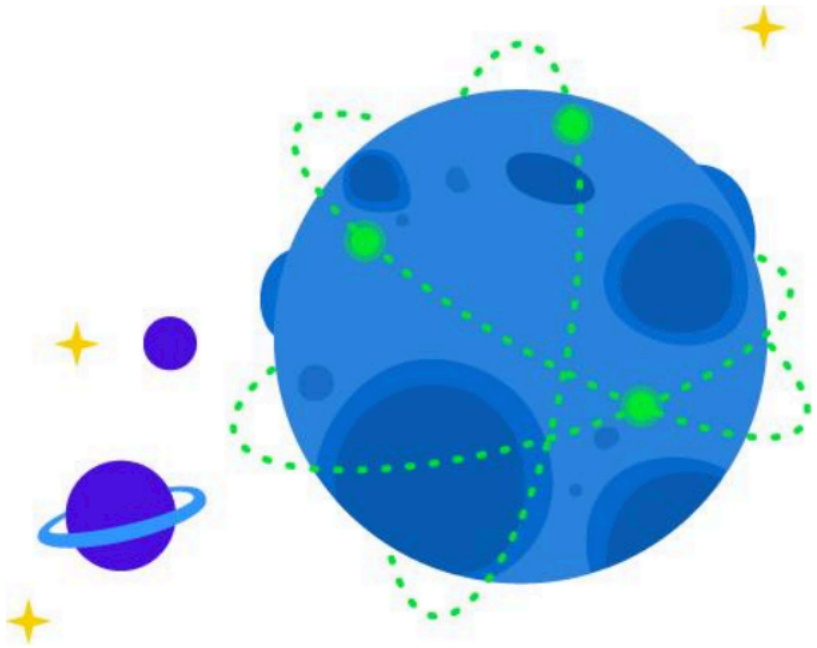
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